



Food and Agriculture Organization
of the United Nations

Statistical Standard Series

Observation Status Code List Version 4

Endorsed by the Technical Task Force of the Data Coordination
Group (DCG-T)

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This statistical standard provides the list of recommended observation status codes (flags) to be used when disseminating FAO statistics to external data users. It includes the definitions and guidance on how to use the codes.

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ACRONYMS

CCSA Committee for the Coordination of Statistical Activities

SDMX Statistical Data and Metadata eXchange

SDW Statistical Data Warehouse

SWS Statistical Working System

BACKGROUND

FAO statistical activities aim at producing and disseminating statistics on food, nutrition, agriculture, forestry and fishery that allow comparisons over time and between geographical areas. A large part of FAO statistics is based on secondary data, i.e. data supplied by national statistical authorities or by other international organizations. These data are disseminated by FAO either in their initial form or after some transformation, reclassification or other manipulations, including further aggregation to derive regional and global aggregates. FAO also collects microdata (primary data sources) which serve as the basis for the calculation of estimates on relevant phenomena. In other cases, different types of data are integrated and become the input of complex computations and models aimed at producing estimates, nowcasts and forecasts.

A unified and standardized system of codes (also named flags) to be associated to data disseminated externally by FAO was established in 2016, in order to increase the accessibility and clarity of FAO statistics (Principle 16 of the FAO Statistics and Data Quality Assurance Framework) and provide users with information on the status of the figures (or missing figures) disseminated by the Organization. The proposed system of flags was inspired by the SDMX guidelines on the cross-domain “Code list for Observation Status”, the guidelines on the possible ways of implementing the code list of Observation Status¹ and the guidelines for the creation and management of SDMX Code Lists.² The system was further refined and updated in 2023 and 2025 to better account for the specific characteristics of FAO integrated statistical system and consider the updates of the SDMX Code list for Observation Status (2019; version 2.2).³

The current FAO Observation Status Code list has been aligned with the corresponding SDMX cross-domain code list to the extent possible. However, it was deemed necessary to maintain a few minor differences with respect to the reference system, to account for the specificities of FAO statistical production processes. Annex 2 of this document provides a complete mapping between the code list for observation status established by FAO (and its subsequent revisions) and that recommended by the guidelines of the SDMX Statistical Working Group.

The system of flags outlined in this statistical standard is intended to be used as cross-domains structural metadata by all potential users of FAO statistics, including for data exchange with international organizations using FAO data as input for their statistical production processes.

1. TECHNICAL TERMS AND DEFINITIONS

- **Observation status (flag):** “attribute of a cell in a dataset representing qualitative information on the value of that cell”,⁴ it provides information on the quality of a value or unusual or missing

¹ https://sdmx.org/wp-content/uploads/OBS_STATUS_Implementation_2_0.docx

² https://sdmx.org/wp-content/uploads/Guidelines_on_versioning_v1_0.pdf

³ Code list for Observation Status (Version 2.2 - 2019) https://sdmx.org/wp-content/uploads/CL_OBS_STATUS_v2_2.docx

Possible Ways of Implementing the CL_OBS_STATUS Code List (Version 2.0 - 2019): https://sdmx.org/wp-content/uploads/OBS_STATUS_Implementation_2_0.docx

Guidelines for the Creation and Management of SDMX Code List (Version 4.0 - 2025): https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fsdmx.org%2Fwp-content%2Fuploads%2FSDMX_Guidelines_for_CDCL-1.docx&wdOrigin=BROWSELINK

⁴ OECD Glossary of Statistical Terms https://www.oecd.org/en/publications/oecd-glossary-of-statistical-terms_9789264055087-en.html

values. In SDMX, flags are translated into codes that describe the *observation status*, where an *observation* is the “value of a variable”.

- **Primary data:** is commonly referred to a source of data (primary data source) collected for statistical purposes, i.e. a set of data referred to statistical units (individuals, households, enterprises, farms, etc.) that represent the basis for the production of statistical outputs (secondary data) on one or more phenomena for the considered target population. Popular examples of primary data sources are represented by the data collected in a Census or a sample survey carried out by a national statistical agency.
- **Secondary data:** in this document it is referred to statistical outputs compiled usually by national statistical agencies based on primary data (see previous definition) in order to get insights on some social, environmental or economic phenomena. The term “secondary” reflects the fact that FAO usually collects these statistical outputs from national institutions (or by another designated sender agency), rather than collecting directly the primary data on which the statistical outputs are based on. Note that the adopted definition of “secondary data” should not be confused with that of “secondary data source” that is often used to identify all data sources other than data collected for statistical purposes (primary data sources).⁵
- **Derived variable:** is a variable that is created as a function of one or more other variables. The creation of a derived variable can involve a simple transformation of a variable (e.g. units of measure conversion), ratio of variables, sums, averages, and more complex computations.
- **Regional and global aggregates:** statistical outputs referred to geographical sub-regions, regions or to the whole world obtained by aggregating the statistical outputs referred to the countries belonging to each target geographical area.

2. SCOPE

This statistical standard presents FAO’s list of cross-domain observation status codes and system of flags, which should be used for the dissemination of FAO’s statistical outputs and for data exchanges with other international organizations.

It applies to all statistical outputs obtained from primary data, secondary data, derived variables, and statistical aggregates disseminated by the Organization and provides the general and technical guidelines for the correct implementation and use of the proposed system of flags.

3. GENERAL RECOMMENDATIONS

- Each statistical output disseminated externally by FAO should come along with the corresponding flag as indicated in the next section on technical recommendations.
- Also missing statistical outputs should have an associated flag explaining the reason for the missing values.
- A statistical output should be associated with only one single flag. As the proposed flags are not always mutually exclusive (e.g. a missing value can generate a break in time series, an estimated value can be of low reliability, data provided by other international organizations are often official

⁵ UNECE (2011) Using Administrative and Secondary Sources for Official Statistics - A Handbook of Principles and Practices, United Nations Economic Commission for Europe
https://unece.org/DAM/stats/publications/Using_Administrative_Sources_Final_for_web.pdf

country estimates, etc.), the flag that occupies a higher position in the hierarchy should be adopted. The hierarchy of flags adopted by FAO is reported in **Annex 1**.

4. TECHNICAL RECOMMENDATIONS

Flags for primary data

It is not necessary to attach an observation flag to microdata disseminated by FAO as the nature of these data (primary data collected by FAO or another organization) would be described in the metadata.⁶

The Flag “E” (Estimated value) should always be used to accompany statistical outputs produced by FAO on the basis of primary data collected, harvested, or obtained by the Organization (e.g., national estimates produced on the basis of a nationally representative survey collected by FAO, statistics calculated from a microdata file publicly available or shared with FAO by a national statistical agency). When estimates derived from primary data are suppressed for confidentiality reasons, the flag “Q” (Missing value suppressed) should be used instead.

Flags for secondary data

Table 1 provides the list of flags that can be assigned to secondary data collected by FAO from national statistical agencies or other organizations and disseminated externally, as well as to missing secondary data.⁷ In this context, national statistical agencies should be considered as the sender agencies, while FAO is intended as the receiving agency.

Table 1 – Observation Status code list associated to secondary data (including missing values) disseminated by FAO

ID	NAME	DESCRIPTION FOR DISSEMINATION	NOTES FOR INTERNAL USE
A	Official value	Value provided as official when the source agency assigns sufficient confidence that it is not expected to be dramatically revised.	This code is the default flag assigned to an official figure that is originally reported to FAO by a statistical agency being part of the national statistical system of a country.
B	Time series break	Observations are characterized as such when different content exists or a different methodology has been applied to this observation as compared with the preceding one (the one given for the previous period).	
E	Estimated value	Observation obtained through an estimation methodology or based on the use of a limited amount of data (e.g. to produce a value at an early stage of the production process while not all data are available).	If needed, additional information on the estimation process and approach can be provided through free text using apposite flag notes at the observation level or at a higher level

⁶ See the Statistical Standard on Microdata Dissemination at <https://openknowledge.fao.org/items/9549963e-c2b2-45a9-b88c-8812fc01d667>

⁷ The table does not apply to derived variables (i.e. statistical outputs that would not exist without the work of FAO) obtained through transformation of primary or secondary data implemented by FAO.

		This code is to be used when the estimation is done by the sender agency (and flagged as such). When the estimation is carried out by a receiver agency in order to replace an outlier or fill a gap in reported data series, the flag to use is I "Value imputed by a receiving agency".	
F	Forecast value	Value deemed to assess the magnitude which a quantity will assume at some future point of time (as distinct from "estimated value" which attempts to assess the magnitude of an already existent quantity).	This code should not be used to flag values referred to moments in time preceding the year of data dissemination, even if these are estimated through forecasting techniques applied on past available data.
G	Experimental value	Data collected on the basis of definitions or (alternative) collection methods under development. Data not of guaranteed quality as normally expected from provider.	
I	Value imputed by a receiving agency	Observation imputed by a receiving agency to replace or fill gaps in reported data series. This code is intended to cover all cases where a receiving agency publishes data about a sending agency that do not come from an official source in the sender agency's reporting framework. When the estimation is done by the sender agency, the flag to use is E "Estimated value".	Cases where imputations are based on data from external unofficial or semi-official data sources are excluded and described under "X" (Value from external organization).
L	Missing value; data exist but were not collected	Used, for example, when some data are not reported/disseminated because they are below a certain threshold.	
M	Missing value; data cannot exist	Used to denote empty cells resulting from the impossibility to collect a statistical value (e.g. a particular crop may not be grown in a given country, making the measurement of its area and production not applicable).	
N	Not significant	Used to indicate a value reported as 0 which is not a "real" zero (e.g. a result of 0.0004 rounded to zero).	
O	Missing value	This code is to be used when the reasons why data are missing cannot be determined.	
P	Provisional value	An observation is characterized as "provisional" when the source agency – while it bases its calculations on its standard production methodology – considers that the data, almost certainly, are expected to be revised.	
Q	Missing value;	Used, for example, when data are suppressed due to statistical confidentiality	Additional information on why observations have been suppressed can be provided through free text

	suppressed	considerations.	using apposite flag notes at the observation level or at a higher level
S	Strike and other special events	Special circumstances (e.g. strike) affecting the observation or causing a missing value.	
U	Low reliability	This indicates existing observations, but for which the user should also be aware of the low quality assigned.	
V	Unvalidated value	Observation as received from the respondent without further evaluation of data quality.	
X	Value from external organization	Observation from an external organization that does not use any system of flags when sharing data. The observation status code "X" can be used for three different types of values: observations from an international/mandated organization, data from unofficial data sources (NGOs, private companies/associations or other non-official/non-commissioned data sources), and mirrored data (historical and current).	The flag "X" should always be complemented by information on the data source through free text using flag notes at the observation level or at a higher level (e.g. in the reference metadata).

Flags for secondary data – Implementation guidelines

Flag A – Official value

The flag "A" is the default flag assigned to an official figure that is originally reported to FAO by a statistical agency being part of the national statistical system of a country.

If the consulted national statistical agency does not indicate a flag or an annotation indicating special circumstances in association to an observation, then the default code "A - official value" should be assumed.

The flag "A" applies also to statistical outputs taken from official country publications or websites.

If a national statistical agency explicitly provides a data point but states that it cannot be considered official, the flag to use should be derived based on the flag of the sender agency (e.g. "P" flagged values should be flagged "P") or if no flag is provided, the default flag to use is "E" (estimated value).

Official data may be collected by *external organizations*; two cases can be distinguished:

- i. Data falling under FAO mandate but that are collected by other international/regional bodies having a formal agreement (e.g. a Memorandum of Understanding, etc.) with FAO (outsourced data collection);
- ii. Data collected from national agencies by another international organization (usually being members of the Committee for the Coordination of Statistical Activities (CCSA)) because the data fall under the mandate of that organization (not under FAO mandate).

Official data collected by bodies listed under item (i) should be flagged as "A - official value" unless otherwise stated; i.e. a value imputed by the organization collecting data on behalf of FAO should take the flag "E – estimated value".

When collecting data from international/mandated organizations (group ii) already adopting the SDMX Code list for Observation Status, a data point flagged as "A" by the sender should be flagged as "A" by FAO. If the data come with no system of flags or without information on the source and the quality of

value, then the flag “X - Value from external organizations” should be used. In this case, additional information on the data source should be provided through a flag descriptor (e.g. a flag note or a reference metadata) as mentioned in Table 1 (section focused on the flag “X”).

Flag B – Time series break

The flag “B” can be associated to both a missing and non-missing disseminated secondary data point.

If a disseminated value comes along with the flag “B”, it indicates that this cannot be compared with the value preceding it in the time series.

A missing value associated with the “B” flag indicates that the given target value either cannot be produced at this time or will no longer be disseminated (e.g. because of a change in concepts or definitions).

Flag E – Estimated value

The flag “E” must be used when a national statistical agency provides a data point but explicitly states that it cannot be considered “official” because, for instance, obtained as the result of a non-standard process.

In statistics, the concept of “**estimation**” is quite general and denotes any “inference about the numerical value of unknown population values from incomplete data such as a sample”;⁸ although following this definition also an official value is the outcome of an estimation process, in this system of flags for secondary data the term “estimation” is used to denote a value obtained applying an estimation methodology different from the one normally used to produce the official figures.

When collecting data from external organizations already adopting the SDMX Code list for Observation Status, a data point flagged as “I” (imputed) or “E” (estimate) by the sender will get the flag “E” by FAO.

Nowcasting cannot be seen as an imputation of a missing value. As such, the “E” flag should also be used to denote **nowcasts** produced by FAO or provided to FAO by another international organization.

Backcasting⁹ is a statistical technique being employed to ensure the coherence of the time series across time. When used by FAO to compensate missing values causing breaks in time series, backcasts should be considered as imputed values and flagged as “I” (note that this is handled in a different manner in SDMX CL_OBS_STATUS v. 2.2, which instead suggests using the flag “E”). However, if backcasts are done by national reporting agencies or by international/supranational organizations providing data to FAO, they should be flagged as “E”.

A missing value replaced by FAO with a non-missing statistical output should be flagged by “I” (imputed value) when it is imputed through a model or a mathematical/computational procedure. When the imputed missing values are collected from international organizations/mandated organizations, non-official sources (through cold-deck imputation), or are obtained as mirrored statistics, the “X” flag should be used instead. In this case, additional information on the data source should be provided through a flag description note.

⁸ OECD glossary of statistical terms; “making an educated guess about a quantity that is unknown, but concerning which some information is available” under “Estimation, classical” in *Encyclopedia of statistical sciences 2nd edition*, edited by N. Balakrishnan et al., John Wiley & Sons, 2006.

⁹ Backcasting refers to forecasting backward in time; the term has also been used for extrapolation. This is done by applying the forecasting method to a series starting from the end and going to the beginning of the data.

<https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Backcasting>

Flag F – Forecast value

The flag “F” should be used to denote **forecasts**¹⁰ done by FAO, or forecasts produced by national statistical agencies and disseminated by FAO.

When collecting data from other international organizations already adopting the SDMX Code list for Observation Status, a data point flagged as “F” (forecast) should also get the flag “F” in FAO database.

The flag “F” can be associated only to values referring to some future points in time. On the other hand, this code should not be used to flag values referred to moments in time preceding the year of data dissemination, even if these have been imputed through forecasting techniques applied on past available data. This means that when forecasting techniques are used to improve the timeliness of time series based on secondary data (i.e. data that are normally received from national focal points or other designated organizations), the outcoming values should be flagged with “I” and not with “F”.

Flag G – Experimental value

Experimental statistics are data obtained using new data sources and methods under development different from that normally used to produce the official values. Being procedures under development, in general it may not be guaranteed quality as normally expected from provider, although the experimental method may be designed to better respond to our users' needs in a timely manner.

When using experimental methods/procedures that have not reached full maturity in terms of harmonization, coverage or methodology, etc., it is suggested to flag the outcoming values with “G – Experimental value”.

Flag I – Imputed value

The flag “I” should be used in all the cases where FAO:

- i) imputes a missing value, i.e. a quantity that exists but could not be collected; or
- ii) replaces a collected value not considered reliable with a value estimated by FAO.

When FAO imputes a missing value with data collected from an external organization that does not adopt the SDMX Code list for Observation Status nor alternative system of flags providing information on source and quality of value, then the imputed values should be flagged as “X”. Additional information on the data sources should be provided through an annotation or through the reference metadata.

When the value imputed by FAO (in replacement of a missing value or an unreliable collected value) is NOT ratified/validated by national agency that was expected to produce it, FAO can disseminate the imputed value flagging it as “I” and adding an annotation explaining that it is not yet ratified/validated by national agency that had to compile and send it to FAO.

In case of a missing value replaced by FAO with a 0 because the phenomenon is assumed negligible for the considered unit, the flag to use is “I” (imputed) and NOT “N – not significant”, usually associated to a 0 obtained as a result of rounding (not being a “real zero”).

The flag “I” should also be used to denote backcasts done by FAO to replace missing values back in the past of the considered time series or to replace past values in time series considered not comparable with the latest ones because of break in the series (due to a change in the definitions/methodology).

When a missing value is related to a phenomenon that does not exist in the country (the measurement cannot be done), the missing value must NOT be replaced by any value (including 0) through imputation. It should be flagged as “M”.

¹⁰ Estimate of a quantity referred to a time reference beyond the dissemination date.

Flag L – Missing value; data exist but were not collected

The flag “L” can only be associated to a missing value when the target quantity exists, but it is not possible to collect/estimate it.

When the missing value provided to FAO is caused by a special event (strike, extreme climatic event, war, etc.) and the missing value is not replaced (imputed) by FAO with a non-missing value then the flag to use is “S – Strike and other special events”.

Flag M – Missing value; data cannot exist

The flag “M” can only be associated to a missing value referring to a target quantity that cannot exist (e.g. the planting area or the production of a crop that cannot be harvested in a given country). In doubt, if it is unknown whether the data is missing because it exists but was not collected (Flag L) or because it is the result of a target quantity that cannot exist, the missing value should be flagged as “O” (see below)

In case of a quantity that cannot exist, the dissemination of a “O” value associated with the Flag “M” or any other flag (as per hierarchy rules) should be avoided. The value should be left missing and not imputed.

Flag N – Not significant

The Flag “N” is used together with a “O” value to indicate a value that is not a "real" zero, but instead the result of a rounding operation.

Flag O – Missing value

The flag “O” can only be associated to a missing value. It must be used when there is no additional information to ascertain why it is missing (e.g. when flags L and M cannot be used with certainty).

Flag P – Provisional value

An observation is characterized as “provisional” (flagged as “P”) when the source agency (including FAO) considers that it is almost certainly expected to be revised.

Flag “P” should be used for data reported to FAO as provisional by a sender agency that is either a national statistical agency or an international or supranational organization that adopts SDMX in flagging the provided value or clearly states that a given value has to be considered as provisional.

Flag Q – Missing value; suppressed

The flag “Q” can only be associated to a missing value when the target quantity exists, but it is suppressed:

- for confidentiality reasons;
- because the value provided to FAO is deemed of insufficient quality and consequently it is deleted and not replaced by any imputed value;
- because the imputation done by FAO to replace a missing value or a collected value found wrong or of poor quality is NOT validated¹¹ by the national statistical agency that provided or that had

¹¹ Although it would be advisable to ask to national statistical agencies to validate all FAO imputations to ensure country ownership and avoid open disagreements on the data published by FAO, this practice would be too burdensome if applied

to provide it and FAO decides not disseminate any value (note that FAO may also decide to disseminate the imputed values flagging it with “I” with the corresponding annotation or to maintain the country value flagging it as “U - low reliability”).

Additional information on why observations have been suppressed can be provided through free text using apposite flag notes at the observation level or at a higher level (e.g. through reference metadata).

Flag S – Strike and other special events

The flag “S” can be used in connection to both observed and missing values:

- It should be associated to a non-missing value provided to FAO by a national statistical agency to highlight an extreme value that actually occurred in the phenomenon being observed because of a special circumstance (e.g. a strike, an extreme climatic event, a war, etc.).
- It can also be associated to a missing value provided to FAO only when the missingness is caused by a special event and the missing value is not replaced (imputed) by FAO with a non-missing value.

Flag U – Low reliability

The flag “U” should be used to denote data received by FAO from a sender agency which is characterized by poor quality.

In the event of a dispute between FAO and national institutions concerning the quality of the country figure, FAO may disseminate the data submitted by the national agency with the flag “U” that indicates the low quality and non-compliance with quality standards.

The flag “U” should not be associated to provisional values, even if by nature they could be characterized as of low accuracy. However, as they are expected to be revised, they should be flagged as “P”.

Flag V – Unvalidated value

The Flag “V” should be used to denote observations received from a national reporting agency or international agency and used/re-disseminated by FAO without carrying out checks to assess whether they are reliable or not.

Flag X – Value from external organizations

The flag “X” should be associated with figures collected by FAO from an external organization (that does not adopt the SDMX Code list for Observation Status nor an alternative system of flags which provides information on the status and the quality of value (see also implementation guidelines of the flag “A”).

Specifically, the flag “X” should be assigned to:

1. Values collected from international organizations/mandated organizations (i.e. an international organization/supranational organization mandated to collect information within the international system, e.g. CCSA members, or on behalf of FAO or its governing bodies through specific agreements),

extensively and therefore it is preferable to apply validation procedures mainly in case of very visible statistical outputs (e.g. SDG indicators).

2. Values collected from unofficial data sources (NGOs, private companies/associations or other non-official/non-commissioned data sources) through what is considered cold-deck imputation (as mentioned in the FAO Statistical Standard Series on Imputation), and
3. Values that are considered mirror data (e.g. values sourced from COMTRADE and Eurostat databases or official tariff line data, when trade data are not available for a specific country).

Considering that these three scenarios represent data sources of different quality level, the flag “X” should always be complemented by an annotation at the observation level providing information on the type of external data source used among the three considered above. In case the same type of data source is used for all data points included in a given FAO dataset, the information on the type of source can also be provided through the reference metadata.

Flags for derived variables

FAO disseminates the collected secondary data either in their initial form or after some transformation, reclassification or other manipulations, which can involve different data items, and result in so-called **derived variables**. Other examples of derived variables are all statistical outputs that FAO produces based on non-statistical data sources – such as geospatial information – retrieved from unofficial suitable data sources.

Examples of simple data manipulations leading to derived variables are:

- Simple transformations (e.g. unit of measure transformation; recoding/re-classification, etc.);
- Calculation of ratios (e.g. yield=production/area harvested of a given product in a country);
- Calculation of sums or averages (e.g. total land area in a country; average income);
- Calculation of proportions (a part in relation to the total, e.g. proportion of forest area in a country).

In other cases, input data items referred to a country are used for deriving complex indicators, as in the case of some SDG indicators (e.g. the country’s Agriculture Orientation Index; the Food Loss/Waste Index, etc.) and more in general variables that cannot be directly measured.

When the output of such transformation/manipulation done by FAO is disseminated to external users then it should come along with the proper observation status to allow its correct use and interpretation. The flag to use depends on the flags associated to the contributing values as well as the type of aggregation formula/method. The main rules are provided in the following section.¹²

Table 2 – Flags that can be associated to the derived variables compiled by FAO

ID	NAME
A	Official value
B	Time series break
E	Estimated value
F	Forecast value
M	Missing value; data cannot exist

¹² To better clarify the type of statistical outputs to which this section of the standard applies, it should be considered that the technical recommendations related to derived variables are relevant to statistics (such as indicators and indexes) that would not exist without FAO’s work on the data.

N	Not significant
O	Missing value
P	Provisional value
Q	Missing value; suppressed
U	Low reliability
X	Value from external organization

Flags for derived variables – Implementation guidelines

Flag A – Official value

With the exceptions of the particular cases listed below, the flag “A” should be associated to values of derived variable compiled by FAO when:

- all contributing items are flagged as official; or
- at least 2/3 of contributing data items are flagged as official and the remaining items can be estimated/imputed with reasonable confidence. It should be noted that, for derived variables obtained as sum or average of secondary data, the contribution to the total must be intended in volume or value and not in terms of the number of data points.
- the derived variable (e.g. a total) is provided to FAO directly from the data provider (as official data) without however providing the contributing data items (in such case, if FAO imputes the contributing data points, they should be flagged as “I”; if some of the contributing data items are assumed negligible and therefore close to 0, FAO should explicitly disseminate a 0 value with the associated flag “I”).

Exceptions/particular cases:

In case of ***simple transformations*** of a single official value, if the transformation follows international standard (e.g. standard unit of measures) then the transformed values should maintain the flag “A”. However, if the transformation is based on rules/coefficients decided by the FAO technical unit and NOT being part of international standards then the corresponding transformed value should be flagged as “E – estimated value”.

In case of ***proportions***, the flag “A” should be used only if the figures at numerator (part of the total amount) and denominator (total amount) are both flagged as “A”.

More generally, for ***ratios***, the resulting figure should be flagged as “A” when both the contributing quantities are flagged as official.

When the derivation of outputs involves ***aggregation*** formulas with coefficients set by FAO according to some assumptions (e.g. weighted sums/averages with weights set by FAO according to some assumptions that are not defined in an international standard), the corresponding statistical output should be flagged as “E” even if all the contributing data entries are all official.

In case of ***sums or simple averages*** where not all contributing items are official, if the values of the non-official items cannot be estimated with a reasonable amount of certainty (and hence they are not imputed) or if they contribute for more than 1/3 of the derived value, the derived variable should be flagged as “E”.

It should be noted that, in cases where the sum or the simple average is equal to 0 it is not possible to apply the decision rule based on the assessment of the contribution of the elements’ value to the total/average. In these cases, the decision rule should be based on the number of observations, i.e. if more than 1/3 is flagged as non-official values, then the derived variable should be flagged as “E”.

Flag B – Time series break

The flag “B” can be associated to both a missing and non-missing derived variable value.

A numerical estimate of a derived variable disseminated with the flag “B” indicates that this cannot be compared with the value preceding it in the time series because of different methods in calculation or in the definition of one or more of the input data points needed to calculate the derived variable. In other words, the flag should be used when there is at least one of the contributing values flagged as “B”.

A missing value associated with the “B” flag indicates estimates referred to a given target quantity that either cannot be produced at this time or will no longer be disseminated (e.g. because of a change in concepts or definitions).

Flag E – Estimated value

The flag “E” has to be used when:

- the computation of the derived variable is done using a complex calculation and/or estimation involving a series of assumptions or coefficients decided by FAO; or
- the final estimate is based on input data points that are flagged in different manners and the flag cannot be derived based on the guidelines related to the other possible flags (other than E) for derived variables discussed in this section.

The flag “E” should also be used to denote nowcasts or backcasts of derived variable values compiled by FAO.

If needed, additional information can be provided through free text using the COMMENT_OBS attribute at the observation level or at a higher level (in a SDMX-compliant environment) or an equivalent flag note.

Flag F – Forecast value

The flag “F” should be used to denote a forecast (a quantity referred to a time reference beyond the dissemination date) for the derived variable produced by FAO. The flag is used for all forecasting exercises, no matter the source of the contributing values (official, imputed, estimated, etc.).

The flag “F” should not be used to flag values of derived variables referred to moments in time preceding the year of data dissemination, even if these have been produced through forecasting techniques applied on past available data. This means that when forecasting techniques are used to improve the timeliness of time series of derived variables, the outcoming values should be flagged with “E” and not with “F”.

Flag M – Missing value; data cannot exist

The flag “M” can only be associated to a missing value when it is impossible to derive an estimate for the targeted derived variable because it cannot exist (e.g. the yield of a product that cannot be harvested in a country). In other terms, a derived variable can be flagged as “M” only if all the contributing data points are flagged as “M”. In situations where the data points present a mix of different flags for missing values (L, M, O, Q), the flag to use is “O” as detailed below.

Flag N – Not significant

The Flag “N” can be used together with a “O” to indicate the value of a derived variable that is:

- Not a "real" zero, but instead the result of a rounding operation.
- Obtained from contributing data points that are all flagged as “N”.

- Obtained from contributing items that are either missing because not applicable (flagged with “M”) or not significant (flagged as “N”).

Flag O – Missing value

The flag “O” can only be associated to a missing value. It must be used when it is not possible to compile the derived variable value for different reasons; e.g. all of the input values necessary to compile the derived variable are missing due to various reasons (i.e. the contributing data points present a mix of different flags for missing values – L, M, O, Q); or not all input values are missing, but the combined result cannot be calculated (e.g. missing value for the denominator in a ratio).

Flag P – Provisional value

The flag “P” should be associated to the value of a derived variable compiled by FAO that is completely based on values flagged as “P”.

Flag Q – Missing value; suppressed

The flag “Q” should be associated to a missing value of a derived variable compiled by FAO that is either:

- completely based on elementary missing values flagged as “Q”;
- suppressed because considered confidential or deemed of insufficient quality.

Flag U – Low reliability

The flag “U” should be associated to the value of a derived variable compiled by FAO that is completely based on elementary values flagged as “U”.

Flag X – Value from external organizations

This flag “X” should be associated to the value of a derived variables compiled by FAO that is completely based on values flagged as “X”, with some exceptions that are listed below.

In case of **simple transformations** of a single value flagged as “X”, if the transformation follows international standard (e.g. standard unit of measures) then the transformed values should maintain the flag “X”. However, if the transformation is based on rules/coefficients decided by the FAO technical unit and NOT being part of international standards then the corresponding transformed value should be flagged as “E – estimated value”.

When the derivation of outputs involves **complex aggregation formulas** with coefficients set by FAO according to some assumptions then the corresponding statistical output should be flagged as “E” even if all contributing data entries are all flagged as “X”.

Considering that the flag X can be used to indicate values derived from observations gathered from mandated organizations, unofficial sources, or mirrored data the value of a derived variable compiled by FAO can be considered as:

- coming from a mandated organization only if it is completely based on values coming from mandated organizations.
- obtained as transformation of mirrored data only if it is completely based on mirrored values.
- coming from a non-official source in all circumstances where all or part of the elementary values used to produce the derived variable have been retrieved from a non-official source.

Flags for aggregates

In producing its statistical outputs, the FAO enriches the collected data also by estimating regional and global aggregates for both the collected data and the derived variables. Analogously, secondary and derived data referred to specific categories or items are usually aggregated following the hierarchy established by, for example, adopted statistical classifications. Aggregated statistical outputs disseminated externally by FAO should come along with flags of observation status that should be assigned depending on the input data points and on the aggregation formula (see following section for more details).

Table 3 – Flags that can be associated to statistical aggregates compiled by FAO

ID	NAME
A	Official value
B	Time series break
E	Estimated value
F	Forecast value
M	Missing value; data cannot exist
N	Not significant
O	Missing value
P	Provisional value
Q	Missing value; suppressed
U	Low reliability
X	Value from external organizations

Flags for aggregates – Implementation guidelines

Flag A – Official value

With the exceptions of the particular cases listed below, the flag “A” should be associated to aggregates compiled by FAO when:

- all contributing items are based on official values; or
- at least 2/3 of the contributing data items are official values, and the non-official items can be estimated/imputed with reasonable confidence. The contribution must be intended in volume or value and not in terms of the number of data points.

Exceptions/particular cases

In cases where not all contributing items are official, if the values of the non-official items cannot be imputed with a reasonable amount of certainty or if they contribute for more than 1/3 of the aggregate, the derived aggregate should be flagged as “E”.

It should be noted that, in cases where the aggregate is obtained by summing or averaging the contributing values and the adopted procedure leads to an aggregate equal to 0, the rule based on the assessment of the contribution of the element’s value to the total cannot be applied. In this case, the 1/3 rule should be applied to the number of observations instead, i.e. if more than 1/3 of the contributing observations are flagged as non-official values, then the derived aggregate should be flagged as “E”.

If the derivation of statistical aggregates is based on **aggregation methods** decided by FAO and not being

established as international standards (e.g. a regional aggregate is obtained by applying the weighted average but with a different weighting system other than that suggested in the corresponding international standards/guideline) or when the aggregate is obtained by applying an ad hoc statistical method instead of the standard aggregation formula then the corresponding output value should be flagged as “E” even if all the contributing data entries are all official.

Flag B – Time series break

The flag “B” is associated to both missing and non-missing values of statistical aggregates when one of the contributing values is flagged as “B” or there is a change in the methodology for deriving the aggregate.

A missing value associated with the “B” flag indicates that the estimates referred to a given target quantity that either cannot be produced at this time or will no longer be disseminated (e.g. because of a change in concepts or definitions).

Flag E – Estimated value

The flag “E” has to be used when:

- derivation of statistical aggregates is based on complex aggregation formula involving coefficients set by FAO according to some assumptions (being not established as international standards); or
- the final estimate is based on input data points that are flagged in different manners and the flag cannot be derived based on the criteria listed under the possible flags for aggregates.

The flag “E” should also be used to denote nowcasts or backcasts of values of aggregates compiled by FAO.

If needed, additional information can be provided through free text using the COMMENT_OBS attribute at the observation level or at a higher level (in a SDMX-compliant environment) or an equivalent flag note.

Flag F – Forecast value

The flag “F” should be used to denote a forecasted statistical aggregate (i.e. quantities referred to a time reference beyond the dissemination date).

The flag “F” should not be used to flag values of aggregates referred to moments in time preceding the year of data dissemination, even if these have been produced through forecasting techniques applied on past available data. This means that when forecasting techniques are used to improve the timeliness of statistical aggregates, the outcoming values should be flagged with “E” and not with “F”.

Flag M – Missing value; data cannot exist

The flag “M” can only be associated to a missing value when it is impossible to derive an estimate of the global/regional aggregate because it cannot exist (e.g. the yield of a product that cannot be harvested in a region). In other terms, a statistical aggregated can be flagged as “M” only if all the contributing data points are flagged as “M”. In situations where the data points present a mix of different flags for missing values (L, M, O, Q), the flag to use is “O” as detailed below.

Flag N – Not significant

The Flag “N” can be used together with a “0” to indicate the value of a statistical aggregate that is:

- Not a "real" zero, but instead the result of a rounding operation.
- Obtained from contributing data points that are all flagged as “N”.
- Obtained from contributing items that are either missing because not applicable (flagged with “M”) or not significant (flagged as “N”).

Flag O – Missing value

The flag “O” can only be associated to a missing value. It must be used when there is no additional information to ascertain the reasons for missingness or when target quantity exists but all or some of the points necessary to compile the aggregate are missing due to various reasons (i.e. the contributing data points present a mix of different flags for missing values – L, M, O, Q) and they are not imputed before aggregation.

Flag P – Provisional value

The flag “P” should be associated to the value of a statistical aggregate compiled by FAO that is completely based on values flagged as “P”.

This flag should also be used when FAO decides to produce aggregates on the basis of a limited number of contributing values (e.g. when data collection in some countries is still ongoing, or only a limited set of countries have reported data to FAO).

Flag Q – Missing value; suppressed

The flag “Q” should be associated to a missing value of a statistical aggregate compiled by FAO that is either:

- completely based on elementary missing values flagged as “Q”;
- suppressed because considered confidential or deemed of insufficient quality.

Flag U – Low reliability

The flag “U” should be associated to the value of a statistical aggregate compiled by FAO that is completely based on elementary values flagged as “U”.

Flag X – Value from external organizations

The flag “X” should be associated to statistical aggregates compiled by FAO that are completely based on values flagged as “X”.

When the derivation of statistical aggregates is based on **complex aggregation formulas** with coefficients set by FAO according to some assumptions then the corresponding statistical output should be flagged as “E” even if all the contributing data entries are all flagged as “X”.

Considering that the flag X can be used to indicate aggregates based on observations gathered from mandated organizations, unofficial sources, or mirrored data, the value of an aggregate compiled by FAO can be considered as:

- coming from a mandated organization only if it is completely based on values coming from mandated organizations.
- Based on mirrored data only if it is completely based on mirrored data.
- coming from a non-official source in all circumstances where all or part of the elementary values used to produce the aggregate have been retrieved from a non-official source.

5. GOVERNANCE PROCEDURES

- All technical units processing or disseminating data are responsible for the implementation of this Statistical Standard and its subsequent revisions.
- Proposals for extensions or modifications to the standard shall be submitted for approval to the DCG-T and the Chief Statistician. The Data Quality Unit, under the oversight of the Chief Statistician and in close collaboration with DCG-T members, will be responsible for updating this standard accordingly and will document the different versions of the document in an annex.
- The SDMX Statistical Working Group is consulted to align with the cross-domain code list for Observation Status,¹³ to the extent possible.

¹³ https://sdmx.org/?sdmx_news=new-version-of-code-list-for-observation-status-version-2-2

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Annex 1: Hierarchy of FAO flags

The proposed flags are not always mutually exclusive (e.g. a missing value can generate a break in time series, an estimated value can be of low reliability, data provided by other international organizations are often official country estimates, etc.). For this reason and because only one flag can be assigned per observation, the hierarchy presented below is intended to facilitate the selection of flags. The flag having the highest priority (appearing in the top of the table) should be picked-up. Implementation guidelines for statistical processes managed through the FAO Statistical Working System are provided at <https://sdhc.review.fao.org/confluence/x/OwAbB> (internal link).

Observation status hierarchy <i>(from highest to lowest importance)</i>	Relevant in conjunction with...	
	numeric values	missing values
B / time series break (highest importance)	Yes	Yes
O / missing value (unknown reasons)		Yes
M / missing value; data cannot exist		Yes
L / missing value; data exist but were not collected		Yes
Q / missing value; suppressed		Yes
S / strike and other special events	Yes	Yes
I / imputed value	Yes	
F / forecast value	Yes	
E / estimated value	Yes	
P / provisional value	Yes	
N / not significant	Yes	
U / low reliability	Yes	
V / unvalidated value	Yes	
G / experimental value	Yes	
X / value from external organizations	Yes	
A / official value	Yes	

Annex 2: Mapping between the code list for observation status established by FAO and the guidelines of the SDMX Statistical Working Group

This Annex provides an overview of how FAO system of flags evolved over time, as well as a mapping between the FAO and the SDMX code list.

FAO (2016)		FAO (2023)		FAO (2025)		SDMX CL_OBS_STATUS (2019)	
B	time series break	B	time series break	B	time series break	B	time series break (highest importance)
O	missing value	O	missing value (unknown reasons)	O	missing values (unknown reasons)	O	missing value
M	missing value (cannot exist, Not Applicable)	M	missing value; data cannot exist	M	missing value; data cannot exist	M	missing value; data cannot exist
M	missing value (cannot exist, Not Applicable)					H	missing value; holiday or weekend
M	missing value (cannot exist, Not Applicable)	L	missing value; data exist but were not collected	L	missing value; data exist but were not collected	L	missing value; data exist but were not collected
Q	missing value; suppressed	Q	missing value; suppressed	Q	missing value; suppressed	Q	missing value; suppressed
						J	derogation
S	exceptional event	S	strike and other special events	S	strike and other special events	S	strike and other special events
						D	definition differs
						K	data included in another category
						W	Includes data from another category
I	imputed value	I	imputed value	I	imputed value	I	imputed value
F	forecast value	F	forecast value	F	forecast value	F	forecast value
E	estimated value	E	estimated value	E	estimated value	E	estimated value
E	estimated value	G	experimental value	G	experimental value	G	experimental value
E	estimated value	U	low reliability	U	low reliability	U	low reliability
P	provisional value	P	provisional value	P	provisional value	P	provisional value
N	not significant (negligible)	N	not significant	N	not significant	N	not significant
		V	unvalidated value	V	unvalidated value	V	unvalidated value
<blank>	official figure	A	official value	A	official value	A	normal value
R	revised official figure						
T	unofficial figure						
X	figure from international organization	X	value from international/mandated organization	X	Value from external organizations		